

Summary sheet

MDS plots

	group	...
s01	drugA	...
s02	drugB	...
s03	drugC	...

sample info

MDS plots show relationships between samples (closer together = more similar). Make one with `plotMDS(dge)` and you hope to see clear separation by a biologically meaningful variable:



Design matrix `model.matrix()`

Describes the covariates

Rows = samples, columns = groups

	groupBasal	groupLP	groupML
10_6_5_11	0	1	0
9_6_5_11	0	0	1
purep53	1	0	0
JMS8-2	1	0	0
JMS8-3	0	0	1
JMS8-4	0	1	0
JMS8-5	1	0	0
JMS9-P7c	0	0	1
JMS9-P8c	0	1	0

Value of 1 indicates sample is in that group, 0 indicates it is not

Contrast matrix `makeContrasts()`

Describes the comparisons

Rows = groups, columns = comparisons

Contrasts			
Levels	BasalvsLP	BasalvsML	LPvsML
Basal	1	1	0
LP	-1	0	1
ML	0	-1	-1

Pay attention to the order of the comparison (e.g. BasalvsLP gives a +ve logFC if gene is higher in Basal)

`voom()` transforms RNA-seq data to allow use of a linear model. Use it like:

```
v <- voom(dge,
  design, plot = TRUE)
```

Where `design` is your design matrix, and `plot = TRUE` makes the mean-variance (voom) plot

```
vfit <- lmFit(v)
```

Find DE (differentially expressed) genes
using Empirical Bayes modified t-tests
with the eBayes() function:

Then, **perform the tests:**

```
dt <- decideTests(efit), and find  
the top 100 DE genes for "Basal vs LP"  
using topTable(  
  efit, coef = "BasalvsLP",  
  n = 100)
```

Visualising DE With volcano plot

